

Analysis of Integral Membrane Protein–Antibody and Lipid-Protein Interactions Using the ProteOn™ XPR36 Protein Interaction Array System

Deborah J. Tew², Moran Tabul¹, Laura Moriarty¹, and Tsafir Bravman¹
¹Bio-Rad Laboratories, Inc., 2000 Alfred Nobel Drive, Hercules CA 94547; ²Bio21 Molecular Science and Biotechnology Institute, 30 Flemington Road, The University of Melbourne, Parkville, Victoria, 3010, Australia

BIO-RAD

Life Science Group
2000 Alfred Nobel Drive
Hercules, CA 94547 USA

Introduction

The interplay of interactions between membranes, membrane proteins, and antibodies is of great interest in the field of drug discovery and development. Membrane proteins and membrane-associated proteins and peptides have been shown to be important in many biological processes and are involved in key pathways in diseases such as cancer. Membrane proteins are fast becoming the next big class of drug targets.

This poster describes the use of a ProteOn XPR36 protein interaction array system, which uses surface plasmon resonance to measure 36 biomolecular interactions simultaneously, to study the interaction of artificial membranes and two significant peptides. Amyloid beta (A β) peptide is implicated in the pathology of Alzheimer's disease, and α -synuclein peptide is implicated in the pathology of Parkinson's disease. The kinetics of the interaction of A β peptide with the artificial membrane were also determined; these studies were carried out using a single modified ProteOn sensor chip. We also describe the use of a ProteOn NLC sensor chip to capture biotinylated and nonbiotinylated liposomes containing membrane proteins. The dose-dependent interaction of the captured incorporated membrane protein with its corresponding antibody is also shown.

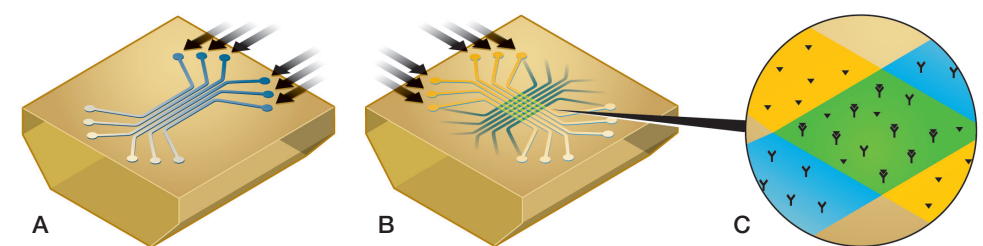


Fig. 1. Generation of the 6 x 6 ligand-analyte interaction array. **A**, six ligands are immobilized in six parallel ligand channels (blue); **B**, six analyte samples are injected into six analyte channels perpendicular to the six ligand channels (yellow); **C**, detail of a single ligand-analyte interaction spot (green) showing the positions of the two interspot references (yellow).

Experiment 1. Modification of a GLM Sensor Chip with Small Unilamellar Vesicles (SUVs) and Subsequent Capture of A β and α -Synuclein Peptides

Chip Modification

The sensor chip surface was pretreated with two pulses each of 50 mM NaOH and 8 mM CHAPS in both the vertical and horizontal directions, followed by three injections of PBS at a flow rate of 100 μ /min in the vertical direction. Channels 1, 3, and 5 were activated with 0.05 M N-hydroxysulfosuccinimide (sulfo-NHS) and 0.02M 1-ethyl-3-[dimethylaminopropyl] carbodiimide HCl (EDAC) at a flow rate of 30 μ /min for 6 min. Undecylamine was prepared by mixing 5 μ l undecylamine with 495 μ l dimethyl sulfoxide (DMSO) and further diluted 20-fold in sodium acetate buffer at pH 5.0. The solution was mixed with a pipet tip until clear. Undecylamine (0.05% v/v) was injected into channels 1, 3, and 5 for 6 min at 30 μ /min; deactivation was performed in channels 1, 3, and 5 with two injections of 1 M ethanolamine for 6 min at 30 μ /min. The surface was stabilized with a single injection of 50 mM NaOH in the vertical direction.

Lipid Binding

SUVs were prepared as described previously (Smith et al., 2008) and diluted to a final concentration of 0.4 mM in PBS. SUVs were injected for 400 sec at a flow rate of 30 μ /min in the vertical direction. Channels 1 and 2 received 1-palmitoyl-2-oleoyl-sn-glycero-3-[phospho-L-choline] (POPC) (0.4 mg/ml); channels 3 and 4 received 1-palmitoyl-2-oleoyl-sn-glycero-3-[phospho-L-serine] (POPS); and channels 5 and 6 received a mixture of POPC and POPS.

Analyte Injection

A β (100 μ g) was prepared as described previously (Tew et al., 2008) to a final concentration of 22 μ M in PBS. Prior to analyte injection in the horizontal direction, the surface was stabilized using three or four injections of PBS also in the horizontal direction at a rate of 100 μ /min for 30 sec. Analytes A β (8.8 μ M) or α -synuclein (7 μ M) were injected in the horizontal direction at a flow rate of 100 μ /min. Association was monitored for 1 min and dissociation for 10 min. For kinetic analysis, a serial dilution of A β (11 μ M, 5.5 μ M, 2.75 μ M, and 1.375 μ M) was injected in four channels, with only buffer injected in the remaining channels.

Chip Surface Regeneration

To regenerate the surface and remove the lipids, two short pulses of 50 mM NaOH and then 8mM CHAPS were injected in the vertical direction for 18 sec at the high flow rate of 100 μ /min.

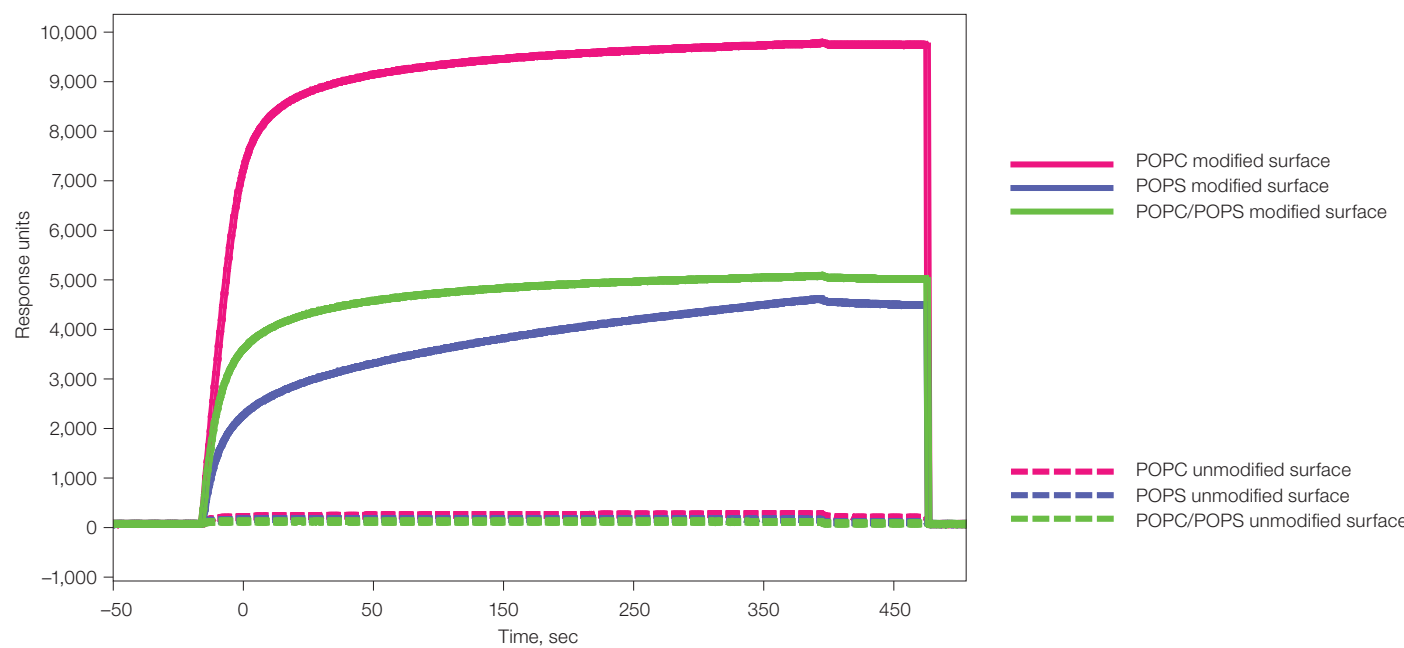


Fig. 2. Specific immobilization of the lipids POPC, POPS, and POPC/POPS to the undecylamine-treated channels of the GLM sensor chip.

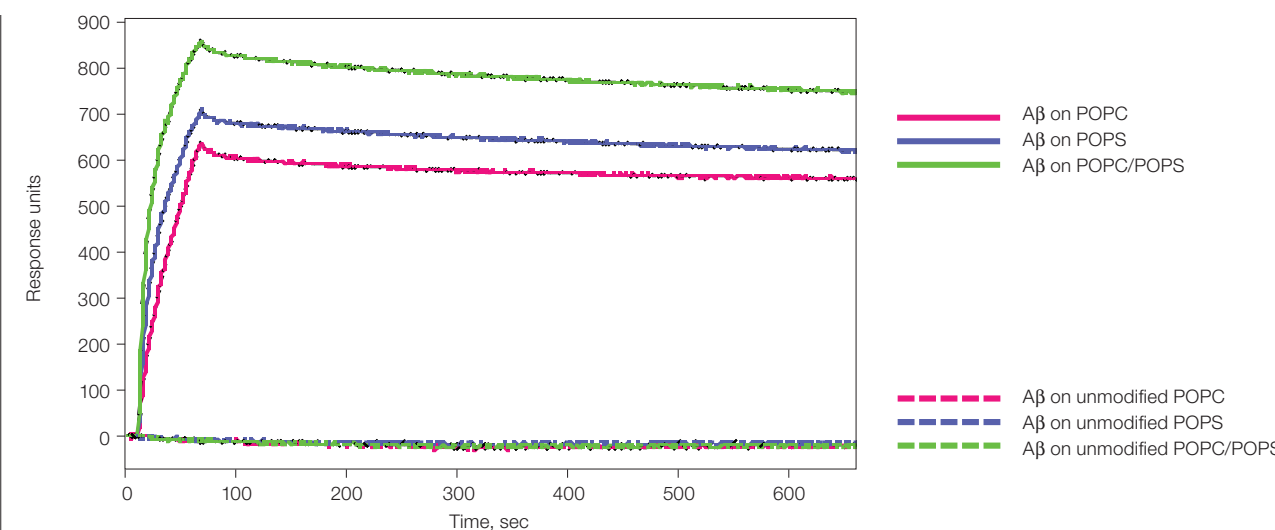


Fig. 3. A β is selectively captured only in the channels that were treated with undecylamine and POPC, POPS, or POPC/POPS lipids; A β is not captured in the unmodified channels.

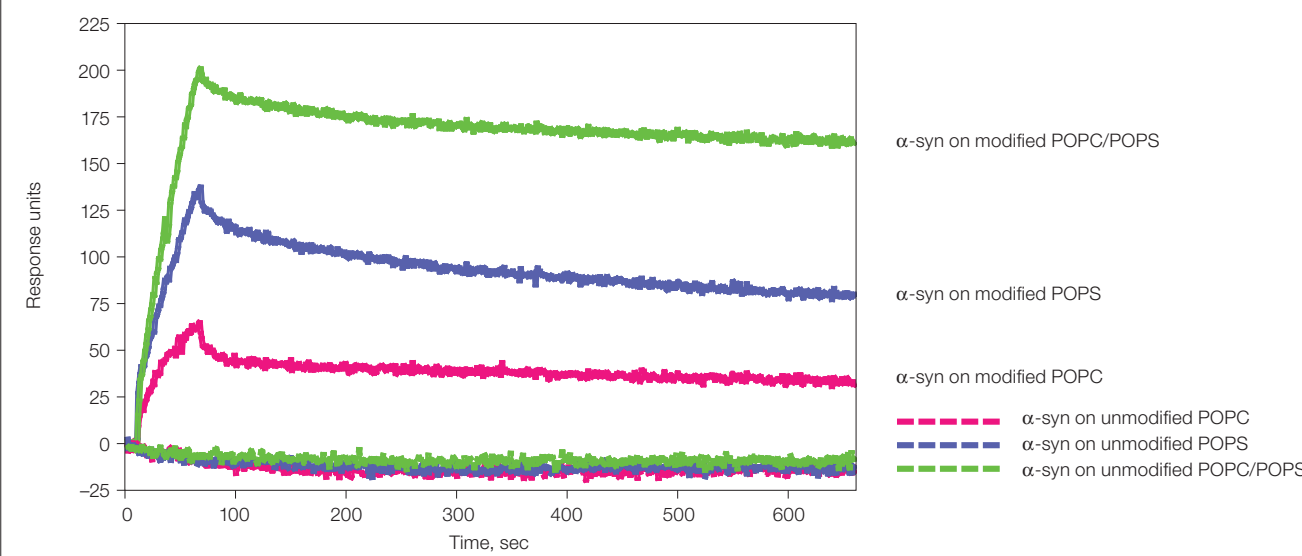


Fig. 4. α -Synuclein is selectively captured only in the channels that were treated with undecylamine and POPC, POPS, and POPC/POPS lipids; α -synuclein is not captured in the unmodified channels.

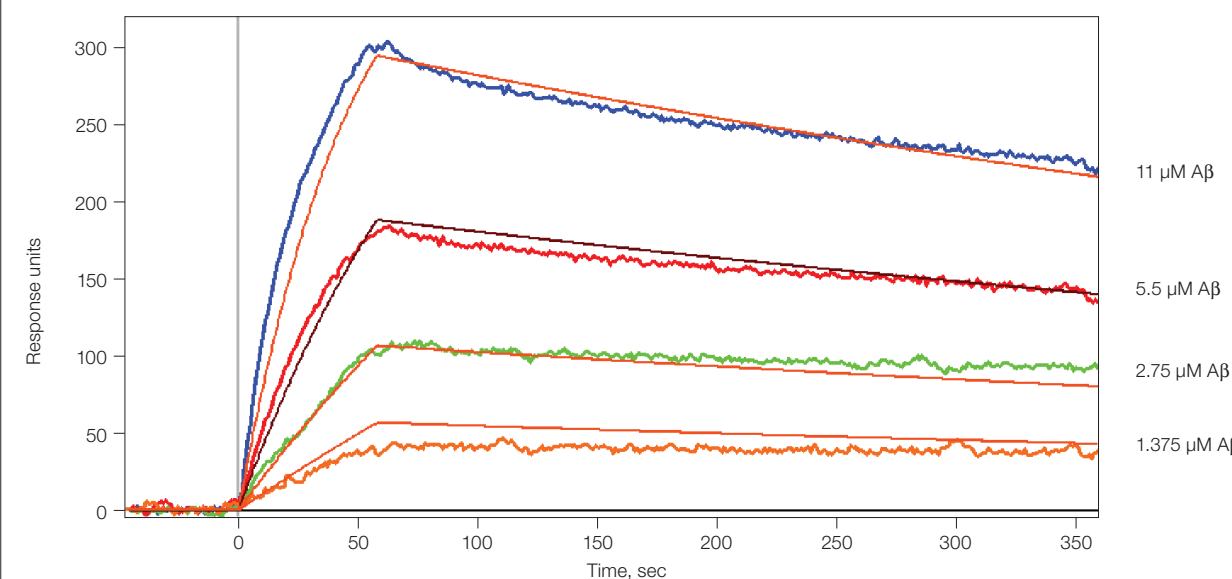


Fig. 5. The dose-dependent interaction of A β with the POPC/POPS modified surface. Kinetic data obtained from the average of A β binding to three lanes: $k_a = 8.5 (\pm 0.3) \times 10^5 \text{ M}^{-1} \text{ s}^{-1}$; $k_d = 3.3 (\pm 0.1) \times 10^{-3} \text{ s}^{-1}$; $K_D = 407 (\pm 9) \text{ nM}$.

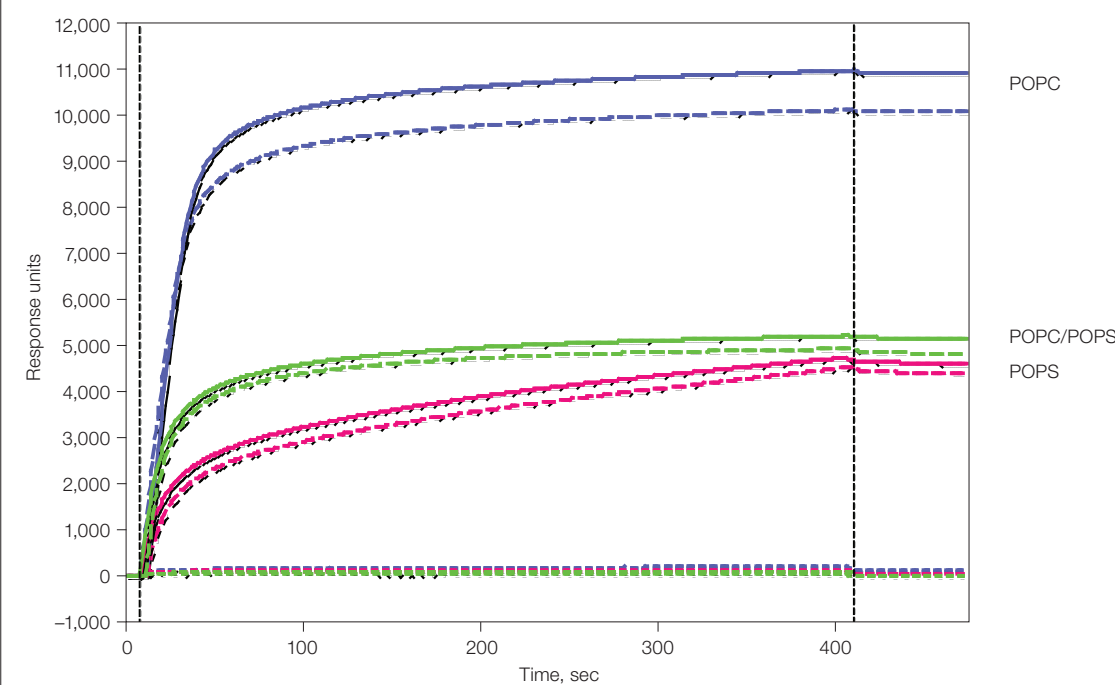


Fig. 6. Recapture of lipids POPC, POPS, and POPC/POPS on the undecylamine-treated GLM sensor chip after regeneration. Channels 1, 3, and 5 were regenerated using 50 mM NaOH and 8mM CHAPS. Solid lines represent first lipid capture on the undecylamine surface; dashed lines represent subsequent capture. The dotted lines represent the lipids injected over the unmodified surfaces in channels 2, 4, and 6.

Experiment 2. Biotinylated Liposome Capture on NLC Sensor Chip Surface

Lipid Binding

Biotinylated dimyristoylphosphatidylcholine (DMPC) liposomes with low level biotinylation containing the herpes simplex virus (HSV)-tagged M2 peptide (a gift from Prof. Shai Arkin, Weizmann Institute, Israel) were diluted 2x in running buffer (PBS), then injected twice for 60 μ l at 25 μ /min into channel 1 in the vertical direction. Nonbiotinylated liposomes and biotinylated liposomes without HSV-M2 peptide were injected in parallel as controls in channels 2 and 3, respectively.

Analyte Injection

A single concentration (333 nM) of the anti-HSV antibodies was injected (120 μ l at 100 μ /min) in the horizontal direction.

Regeneration

The liposomes and antibodies were removed by injection of 20 mM CHAPS for 1.2 min at 100 μ /min and recaptured on the NLC sensor chip for subsequent analysis.

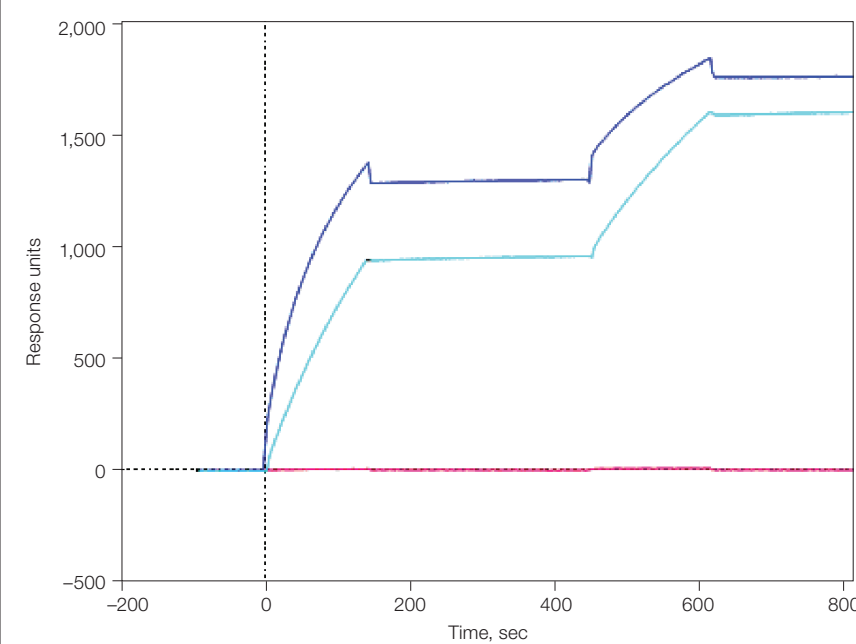


Fig. 7. Biotinylated liposome capture on NLC sensor chips. Channel 1 (blue sensorgram), biotinylated liposomes incorporating HSV-M2 peptide; channel 2 (pink sensorgram), nonbiotinylated liposomes without HSV-M2 peptide (null); channel 3 (light blue sensorgram), biotinylated liposomes without HSV-M2 peptide (null).

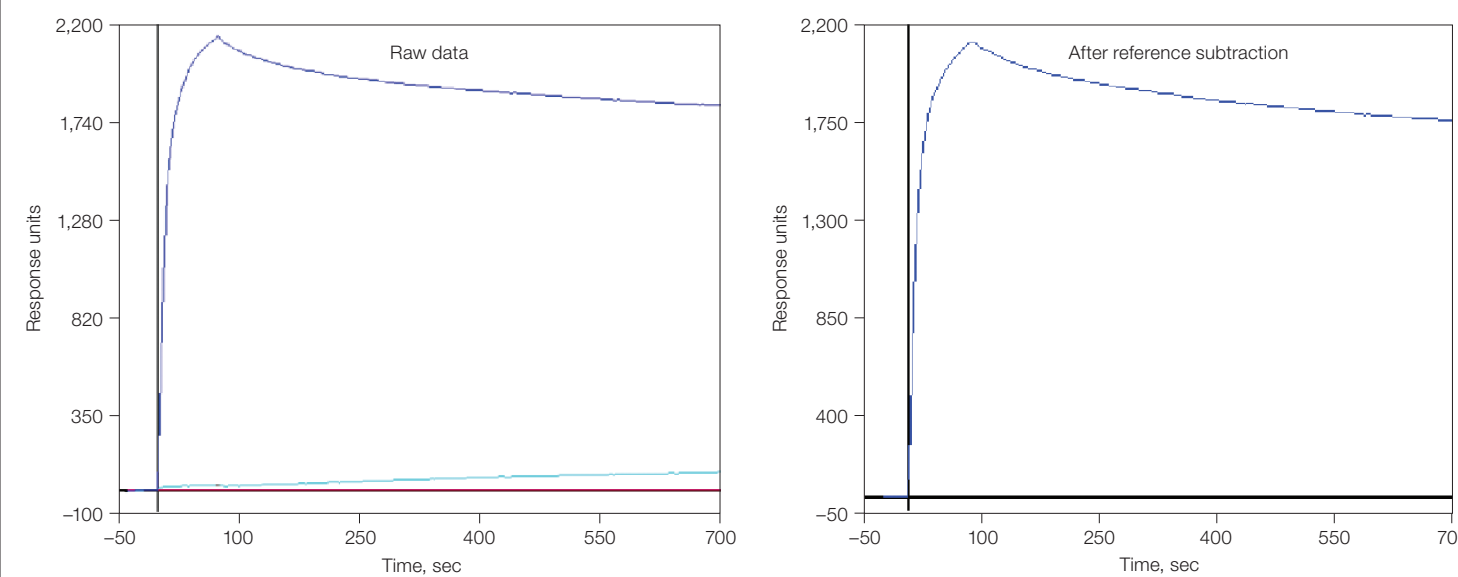


Fig. 8. Anti-HSV antibodies interacting with HSV-M2 peptide incorporated in the biotinylated liposomes captured on an NLC sensor chip. **Left**, anti-HSV antibodies binding to liposomes in channels, raw data. Channel 1 (blue sensorgram), biotinylated liposomes incorporating HSV-M2 peptide; channel 2 (pink sensorgram), nonbiotinylated liposomes without HSV-M2 peptide (null); channel 3 (light blue sensorgram), biotinylated liposomes without HSV-M2 peptide (null). **Right**, specific interaction of anti-HSV antibodies with HSV-M2 peptide incorporated in captured biotinylated liposomes only (referenced against channel 2).

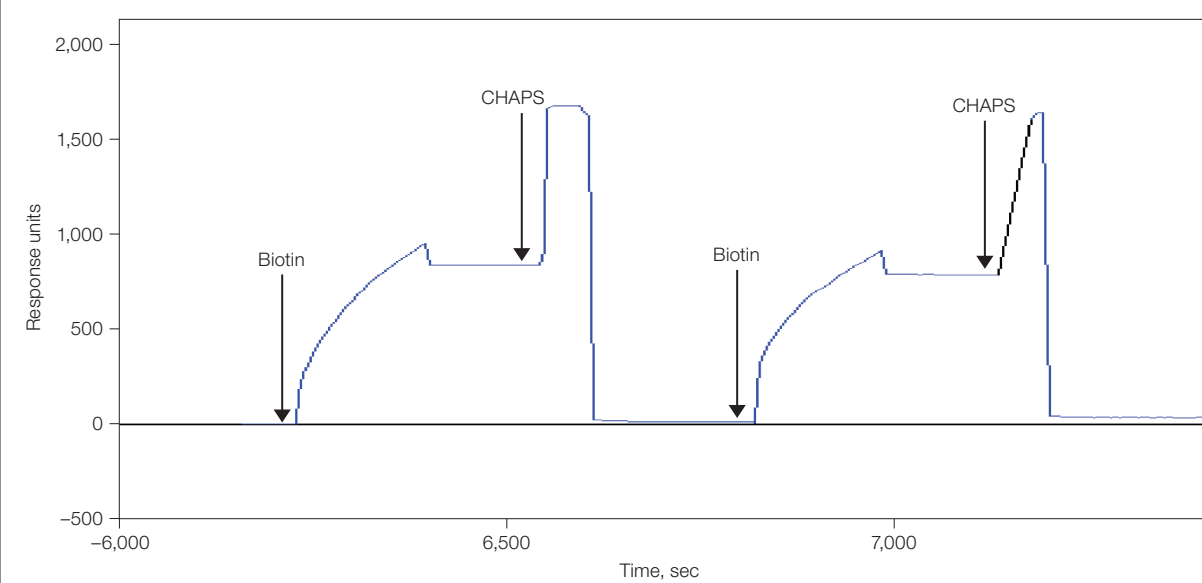


Fig. 9. Liposome regeneration and recapture.

Experiment 3. Nonbiotinylated Liposome Capture on Modified NLC Sensor Chip Surface

Lipid Binding

To avoid creation of vesicles, biotinylated dipalmitoyl phosphatidylethanolamine (DPPE) was diluted in running buffer with 10% DMSO, and 180 μ l was injected into channels 1 and 2 at 25 μ /min in the vertical direction. Nonbiotinylated liposomes containing the HSV-M2 peptide were injected into channel 1 at a rate of 100 μ l at 25 μ /min. Channel 2 was used as a blank reference channel.

Analyte Injection

Three concentrations (10, 5, and 2.5 nM) of anti-HSV antibodies were injected in the horizontal direction (250 μ l each at a rate of 100 μ /min).

Regeneration

The liposomes and antibodies were removed by injection of 20 mM Tween 20 or 20 mM CHAPS (30 μ l at a rate of 25 μ /min).

After regeneration, the liposomes were recaptured onto the NLC sensor chip for subsequent analysis by repeating the injections as described.

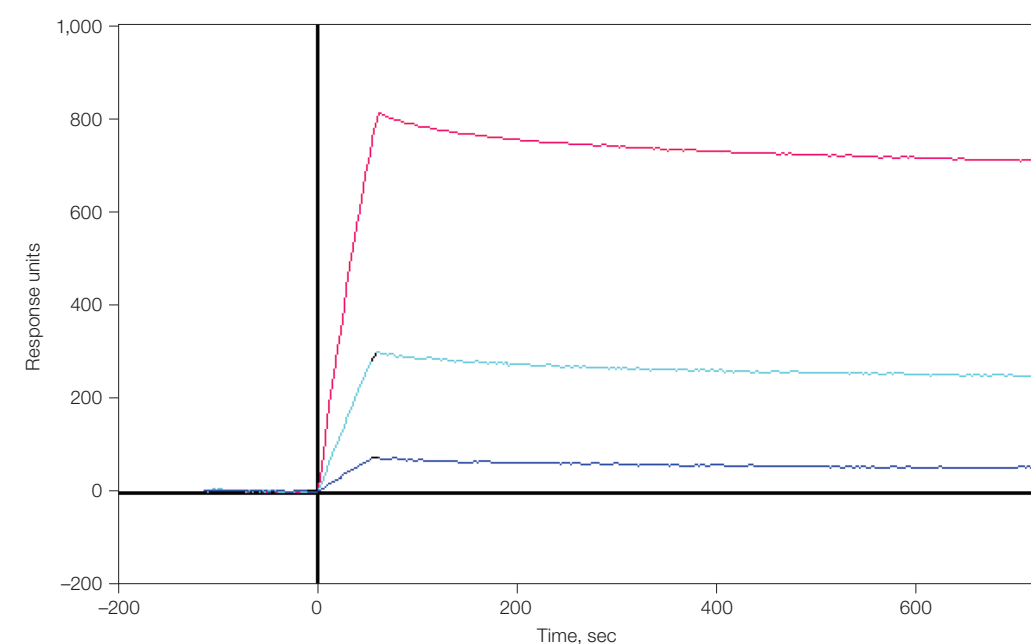


Fig. 10. Anti-HSV antibody (10, 5, and 2.5 nM) interacting with HSV-M2 peptide incorporated in the liposomes captured on the modified NLC sensor chip (a modified channel without liposomes was used as a reference).

Conclusions

- The ProteOn XPR36 protein interaction array system was used to develop easy-to-use, low cost assays to study membrane-protein and membrane protein–antibody interactions
- These interactions can be measured in a high-throughput manner using the ProteOn system's 6 x 6 array
- SUVs made from POPC and POPS were captured reproducibly on a modified ProteOn GLC sensor chip, and the degree of binding of A β peptide and α -synuclein peptide was measured
- The kinetic interaction of the A β peptide with the captured POPC/POPS SUV surface was determined
- Both biotinylated and nonbiotinylated liposomes containing HSV-tagged M2 peptide were captured reproducibly on a ProteOn NLC sensor chip
- The specific interaction of the anti-HSV-tagged antibodies with the M2 peptide incorporated into the liposomes was measured
- The ProteOn protein interaction array system is a powerful tool for developing methods to study the interactions of membranes and membrane-associated proteins, peptides, and antibodies

References

- Smith DP et al. (2008). Formation of a high affinity lipid-binding intermediate during the early aggregation phase of alpha-synuclein. *Biochemistry* 47 (5), 1425–1434.
Tew DJ et al. (2008). Stabilization of neurotoxic soluble beta-sheet-rich conformations of the Alzheimer's disease amyloid-beta peptide. *Biophys J* 94 (7), 2752–2766.
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